sensiva™ sc 50 multifunctional

unique, versatile ingredient for deodorant that improves skin feel and inhibits the growth of odor-causing bacteria

product description

Sensiva™ SC 50 multifunctional is a globally approved, versatile and multifunctional ingredient, as well as a very effective deodorant active. As an emollient, it improves the skin feel of cosmetic formulations. Sensiva™ SC 50 multifunctional reliably inhibits the growth and multiplication of odor-causing bacteria. Additionally, it can boost the efficacy of traditional preservatives and act as anti-microbial stabilizer in combination with other cosmetic ingredients.

origin

Sensiva™ SC 50 multifunctional is a synthetic representative of the 1-alkyl glycerin ethers with a high degree of purity. Substances with a similar structure occur in nature. These alkoxylipids are widely distributed in human and animal tissue. A differentiation is made between neutral and ionic alkoxylipids.

High levels of the neutral alkoxylipids are present in the liver of cartilaginous fish. Batyl alcohol and selachyl alcohol have been isolated from the non-saponifiable proportions of liver oils of sharks and rays. Chimyl alcohol has been found in the liver of the sea rat (Chimaera monstrasa). All of these are 1-alkyl glycerin ethers, namely octadecyl-, hexadecyl- and 9-octadecenyl glycerol. Sensiva™ SC 50 multifunctional is the result of the systematic study of substances in the 1-alkyl glycerin ether class of compounds.

booster for traditional preservative actives

In response to the increasing discussion about preservative actives, there has been interest in reducing the amount of traditional preservatives in cosmetic formulations. Methods of enhancing the efficacy of traditional preservative actives, so that lower levels of these materials can be used, have become increasingly of interest.

A repeated challenge test has been performed to illustrate the use of sensiva™ sc 50 multifunctional to improve the efficacy of traditional cosmetic preservatives, such as phenoxyethanol, benzyl alcohol or methylparaben. The combination of sensiva™ sc 50 multifunctional with phenoxyethanol is offered as easy-to-use blend euxyl™ pe 9010 preservative.

For more information regarding these products please contact your local representative.

key features and benefits

- multifunctional cosmetic ingredient
- skin care additive and emollient
- booster for antimicrobials
- boosting and fixating of fragrance ingredients
- effective against odor-causing bacteria

repeated challenge test (Ashland KoKo Test)

This method is used to determine the preservative effect of chemical preservatives in cosmetic formulations, e.g. creams, lotions and shampoos. For this, in various test series, the preservative to be tested, is added in different concentrations to unpreserved samples. A constant microorganism load is achieved by means of periodic inoculation (inoculation cycles) of the test preparations. Immediately before inoculation samples of the individual preparations are streaked out onto nutrient media. The preservative effect is evaluated on the basis of the microorganism growth on the nutrient media.

The longer the time before the occurrence of the first microbial growth, the more effective is the preservative. Experience has shown that a well preserved product should remain growth-free for six inoculation cycles in order to ensure the shelf-life in the original packaging required in practice (30 months).

sensiva™ sc 50 multifunctional improves the antimicrobial efficacy of traditional preservative actives such as phenoxyethanol, benzyl alcohol and methylparaben

test material / product	inoculation cycles							assessment
	0	1	2	3	4	5	6	
carbomer gel unpreserved	–	+++ B, M	+++ B, M	./.				failed
+ 0.1% sensiva™ sc 50 multifunctional	–	+++ M	+++ B, M	./.				failed
+ 0.9% Phenoxyethanol	–	+++ B	+++ B, Y	./.				failed
+ 1.0% euxyl™ pe 9010 preservative	–	–	–	–	–	–	–	A
+ 0.2% Methylparaben	–	+++ B, Y	+++ B, Y	./.				failed
+ 0.1% sensiva™ sc 50 multifunctional 0.2% Methylparaben	–	–	–	–	–	–	–	A

test material / product alternative o/w cream	inoculation cycles							assessment
	0	1	2	3	4	5	6	
unpreserved	–	+++ B, Y, M	+++ B, Y, M	./.				failed
+ 0.75% Benzyl Alcohol	–	+++ B, Y	+ B, Y	++ B, Y	+ B, Y	++ B, Y	++ B, Y	failed
+ 1.0% Benzyl Alcohol	–	+ B	+ B	+ B, Y	+ B	+ B, Y	+ B	B
+ 0.9% Benzyl Alcohol 0.1% sensiva™ sc 50 multifunctional	–	–	–	–	–	–	–	A

0 = sterility control
B = bacteria
M = moulds
Sp = sporeforming bacteria
Y = yeasts

\- = free of microbial growth
\+ = slight growth
++ = moderate growth
+++ = massive growth

booster for cosmetic alcohols and glycols

Recent discussions about traditional preservative actives has increased interest in novel ways to keep cosmetic products microbiologically stable. It is now more important than ever that preservation or microbiological stability is part of new formulation concepts. Formulators must consider various methods of preventing micro-organisms from growing at an early stage of product development.

Increasing emphasis is being put on the use of multifunctional cosmetic ingredients. These are ingredients which, beside their main benefits, have a certain antimicrobial efficacy, such as glycols, fatty esters or fragrances.

Glycols used at certain levels are known to be antimicrobial. The longer the chain length of a glycol, the lower is the level necessary to prevent micro-organisms from growing. The limitation is the water solubility, as the water solubility decreases with increasing chain length.

Ashland KoKo tests of an oil-in-water lotion have shown that sensiva™ sc 50 multifunctional can improve the antimicrobial efficacy of glycols, such as pentylene glycol or caprylyl glycol. The combination of sensiva™ sc 50 multifunctional with caprylyl glycol is offered by Ashland as easy-to-use blend sensiva™ sc 10 multifunctional.

sensiva™ sc 50 multifunctional improves the antimicrobial efficacy of glycols

test material / product	inoculation cycles							assessment
	0	1	2	3	4	5	6	
o/w lotion unpreserved stabilization	–	+++ B, Y, M	+++ B, M	./.				failed
+ 5.0 % Pentylene Glycol	–	+++ M	+++ M	./.				failed
+ 0.5 % sensiva™ sc 50 multifunctional	–	+++ B, M	+++ B, M	./.				failed
+ 5.0 % Pentylene Glycol 0.5 % sensiva™ sc 50 multifunctional	–	–	–	–	–	–	–	A
+ 0.7 % Caprylyl Glycol	–	+++ B, Y	+++ B, Y, M	./.				failed
+ 0.3 % sensiva™ sc 50 multifunctional	–	+++ B, Y, M	+++ B, Y, M	./.				failed
+ 1.0 % sensiva™ sc 10 multifunctional	–	–	–	–	–	–	–	A

0 = sterility control
B = bacteria
M = moulds
Y = yeasts

\- = free of microbial growth
\+ = slight growth
++ = moderate growth
+++ = massive growth
./. = cancelled

boosting and fixating of perfumes

Fragrances in some products can lose their character rapidly; i.e. the top/middle notes evaporate. It may be desirable to extend the life of such perfumes, particularly with regard to the initial character and intensity of the fragrance.

Two panel tests were conducted to show the influence of sensiva™ sc 50 multifunctional on single fragrance ingredients. The tests evaluated these ingredients on smelling strips and in compounded perfumes in different cosmetic formulations applied to the forearm. The intensity of scent was compared with and without sensiva™ sc 50 multifunctional added to the ingredients or formulations.

A boosting effect is an enhancement of scent by a faster evaporation of the fragrance shortly after application. Fixation is an increase in the lasting power of a fragrance during a prolonged evaporation.

For the first panel test, frequently used fragrance ingredients were selected exemplifying different chemical structures and properties. In the second test, three different perfume compositions were tested in a massage oil, hair gel, eau de toilette, shampoo, oil-in-water emulsion and water-in-oil emulsion.

results of first panel test are listed in the table below:

ingredient name	boosting	fixating	fresher scent	softer scent
cinnamyl alcohol	X	X		
aldehyde C-10 (decanal)	X	X		
aldehyde C-16 (ethyl methylphenylglycidate)	X	X		
citral*			X	
citronellal			X	
hydroxycitronellal*		X		
cinnamal*	X	X		
musk ketone	X	X		
diphenyl ether		X		
citronellol*			X	
linalool*			X	
alpha-terpineol	X	X		
amyl salicylate		X		
anethole			X	
benzyl acetate				X
isobornyl acetate	X	X		
linalyl acetate				X
linalyl isobutyrate	X	X		X
menthanyl acetate				X
methyl anthranilate	X			
caryophyllene	X	X		
amande amère	X	X		
niaouli			X	
patchouli brun huile essence				X

* 76/768/EEC Article 6 (1) (g) substance X = enhancement of the scent by sensiva™ sc 50 multifunctional
Boosting: enhancement of the scent after 1 hour. Fixating: enhancement of the scent after 6 - 24 hours

results

Sensiva™ SC 50 multifunctional had an influence on most of the fragrance ingredients tested with respect to boosting and/or fixation. A fresher or softer appearance of the scent was observed in a few fragrance ingredients after some hours.

Perfume compositions were also boosted and fixated by sensiva™ sc 50 multifunctional in different cosmetic formulations. The lower the substantivity of the perfume, the stronger was the boosting and/or fixating effect of sensiva™ sc 50 multifunctional. In these formulas, the perfume concentration could be reduced with the addition of 0.5 % sensiva™ sc 50 multifunctional.

The boosting and fixating of fragrance ingredients can often allow a lower use concentration of the perfume in the final formulation. Due to interactions between the fragrance ingredients in a perfume composition and with the formulation base itself, the effect of sensiva™ sc 50 multifunctional must be tested case by case.

skin care properties

medium spreading emollient

The selection of emollients is critical to the character of a cosmetic product. High contents of fast spreading emollients lead to light creams with a smooth skin feel. High contents of slow spreading emollients are used in rich creams with refatting properties. Medium spreading emollients close the gap between fast and slow spreading emollients, improving the refatting properties and extending the smooth skin feel of cosmetic formulations. To achieve a long lasting soft and smooth skin feel, a combination of fast, medium and slow spreading emollients is necessary.

Sensiva™ SC 50 multifunctional is a medium spreading emollient with a spreading coefficient of about 700 $mm^2/10\ min$ [1]. The spreading properties are similar to those of dodecyl oleate, hexyldecanol or cetearyl isononanoate.

sensory assessment

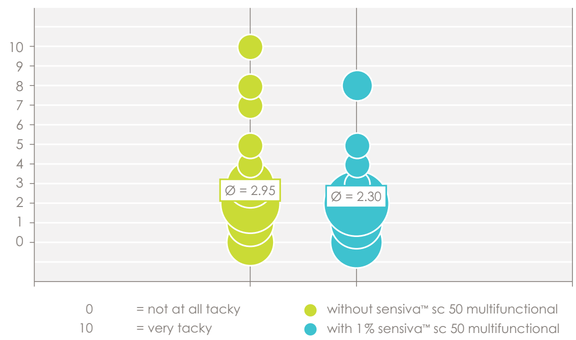
Moisturizing properties play an important role in determining the overall aesthetics of an emulsion. The most frequently used moisturizer in cosmetic formulations is glycerin. However, higher use concentrations of glycerin may lead to some disadvantages in the skin feel of cosmetic formulations, such as slow penetration, tackiness or soaping (whitening during application). A sensory assessment test[2] has been performed on a glycerin-containing oil-in-water emulsion to determine if sensiva™ sc 50 multifunctional can compensate some well-known disadvantages of these formulas.

the following formulations were tested:

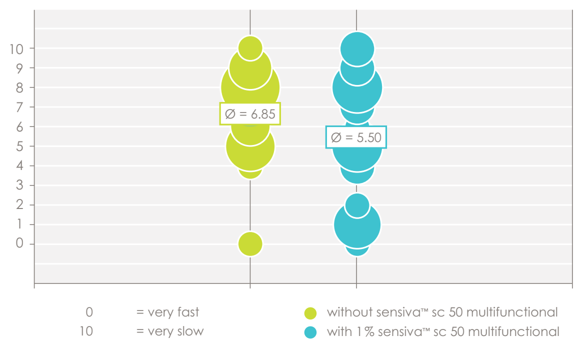
INCI name	cream A	cream B
phase A		
Arachidyl Alcohol, Behenyl Alcohol, Arachidyl Glucoside	5.0	5.0
Cetearyl Ethylhexanoate	20.0	20.0
Ethylhexylgycerin (sensiva™ sc 50 multifunctional)	–	1.0
phase B		
Polyacrylate 13, Polyisobutene, Polysorbate 20	0.2	0.2
phase C		
Aqua	ad 100	ad 100
Glycerin	8.0	8.0
phase D		
preservative	q.s.	q.s.

The sensory assessment test has been carried out on twenty trained subjects using the test material on the inner forearm on a 3 x 3 cm test area. Different aspects such as tackiness, penetration, greasiness, soaping or the overall impression have been evaluated by a ranking of 0 – 10.

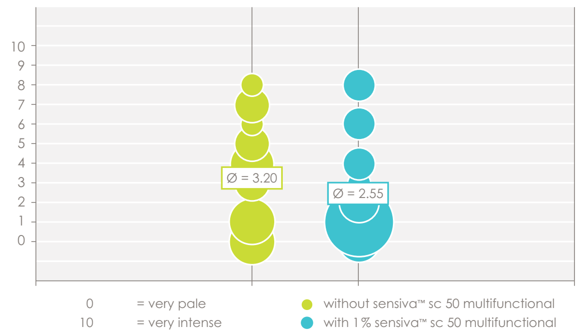
tackiness



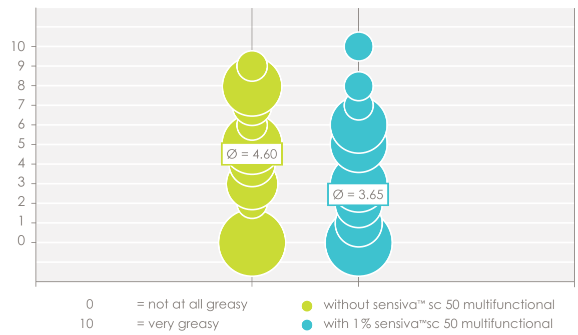
penetration



soaping



greasiness



overall impression

deodorizing efficacy

Body odor arises when sweat, odorless itself, is decomposed by micro-organisms. Gram positive bacteria form substances which have unpleasant odor from the sweat contents.

Sensiva™ SC 50 multifunctional inhibits the growth and multiplication of odor-causing bacteria. A sniff test was conducted to evaluate the efficacy of sensiva™ sc 50 multifunctional in underarm deodorants under practical conditions.

sniff test

A sniff test performed in accordance with the method of Prof. Heiss of the Heidberg Hospital in Hamburg as described in „dragoco report" 6/76[3)], is a study reflecting conditions encountered in practice. It determines the smell-inhibiting effect of the test products by directly sniffing the armpits of test subjects. The test conditions are listed in the table on the next page:

test panel	20 volunteers
sex	female, male
age	> 18 years
test area	armpits
preliminary phase	10 days
duration and frequency of application	twice a day (5 days)
discontinuation of application before rating the initial value	approx. 6 hours after the last wash
discontinuation of application before rating the final value	approx. 6, 16 and 24 hours after the last application of the product
testers	3

In a sniff test, the deodorizing activity of 0.3% sensiva™ sc 50 multifunctional (Deodorant B) was compared with 0.1% triclosan (Deodorant A) as bench mark and also with the basic alcohol/water formulation (Deodorant C) as well. Two sniff tests were performed: Deodorant A versus Deodorant B and Deodorant B versus Deodorant C.

	deodorant A	deodorant B	deodorant C
	% w/w	% w/w	% w/w
Triclosan	0.1	–	–
sensiva™ sc 50 multifunctional	–	0.3	–
Propylene Glycol	1.0	1.0	1.0
alcohol denat.	40.0	40.0	40.0
water demin.	58.9	58.7	59.0

results of sniff test

As can be seen from figures 1 and 2, all three products lead to a clear odor reduction compared to the initial value. Based on figure 1, the Deodorant B – containing sensiva™ sc 50 multifunctional – shows a significantly higher odor reduction compared to the alcohol based product at three evaluation times. The significance of these data was $p < 0.001$ (16 hours after 1 application) and < 0.05 (24 hours after 1 and 12 application).

Based on figure 2, there was no significant difference between deodorants with sensiva™ sc 50 multifunctional and triclosan. The sniff test proves a quantitatively similar smell-inhibiting effect of both test products.

figure 1: Sniff test results deodorant B (with sensiva™ sc 50 multifunctional) versus deodorant C (basic formulation). Values marked with an asterisk show significant differences between both deodorants.

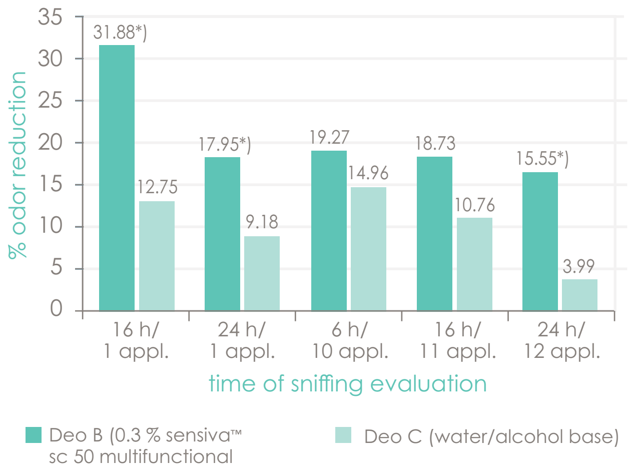
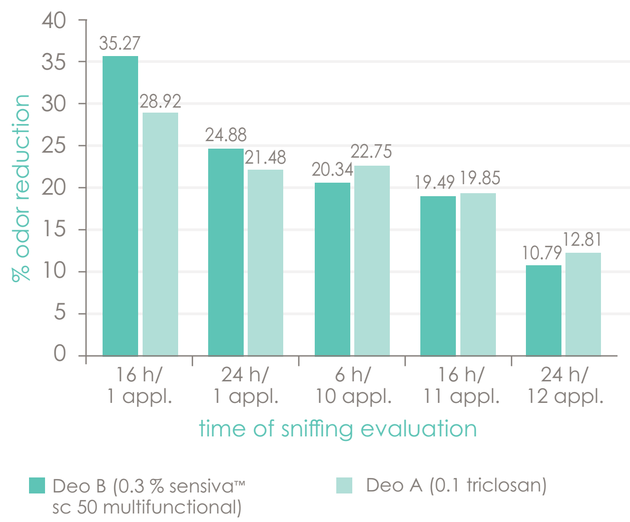


figure 2: Sniff test results deodorant A (with triclosan) versus deodorant B (with sensiva™ sc 50 multifunctional).



global approvals

	leave-on	rinse-off
Ashland recommendation	0.3 – 1.0%	-
EU Cosmetics Regulation	no restrictions for the use in cosmetics	
CIR (USA)	max. 2.0%	max. 8.0%
ASEAN Cosmetic Directive	no restrictions for the use in cosmetics	
TSSC 2015 (China)		
MERCOSUR		

For approvals in further countries please contact your local sales representative

The individual use concentration is dependent on the sensitivity of the product to microbial contamination, the choice of raw materials and production hygiene. For the determination of the optimum dosage we recommend a preservation test, as the efficacy is highly influenced by the other ingredients of a finished formulation. The staff of Ashland's microbiological laboratories will be pleased to give customers the appropriate support.

indications for use

general

Sensiva™ SC 50 multifunctional is a clear, colorless and nearly odorless liquid. It is stable to hydrolysis, temperature and pH. As a result of its good chemical stability, it is highly compatible with all commonly used cosmetic ingredients.

Whereas glycerin esters are attacked by lipolytic enzymes, this is not the case with sensiva™ sc 50 multifunctional due to its glycerin ether structure. Sensiva™ SC 50 multifunctional is effective in pH-ranges up to 12.

applications

Sensiva™ SC 50 multifunctional can be used as skin care additive and deodorant active in a recommended use concentration of 0.3 – 1.0%.

incorporation

Due to its molecular structure, sensiva™ sc 50 multifunctional can have an influence on stability and/or viscosity of some emulsions. It has a surfactant-like structure which can affect the hydrophilic-lipophilic-balance (HLB) of an emulsion.

The HLB value relates to the hydrophilic and lipophilic properties of an emulsifier and is a first hint of the choice of an emulsifier system. Typically, water-in-oil emulsifiers have HLB values in the range of 3 – 6 while oil-in-water emulsifier are in the range of 8 – 18.

Sensiva™ SC 50 multifunctional has a calculated HLB of approximately 7.5. This is relatively high for the use in water-in-oil emulsions and relatively low for the use in oil-in-water emulsions. Thus it can influence a given emulsifier system in the opposite direction.

Detailed information about the influence of sensiva™ sc 50 multifunctional in emulsions as well as possibilities to restore the stability are summarised in a formulation guideline which can be provided upon request.

product-specific properties

material compatibility

In the material compatibility tests with the concentrate of sensiva™ sc 50 multifunctional, stainless steel, brass, copper, zinc and aluminium, as well as polyethylene (PE), polyoxymethylene (POM), poly-amide (PA), hard polyvinyl chloride (hard PVC), polystyrene (PS), polysulphone (PSU), polycarbonate (PC), polymethylmeth-acrylate (PMMA), polyethylenterephthalate (PET) and acrylnitril-butadienstyrolcopolymer (ABS) proved to be suitable materials for handling the undiluted product. Other non-metallic materials should be checked for their suitability. Polytetrafluoroethylene (PTFE), nitril rubber or isobutene-isoprene rubber or polymethylsiloxane should be preferred as sealing material when handling undiluted sensiva™ sc 50 multifunctional. Other sealing material may show swelling or lead to pronounced discoloration of sensiva™ sc 50 multifunctional. Compatibility has to be proved in each case.

solubility

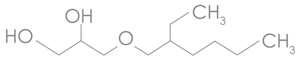
Sensiva™ SC 50 multifunctional has limited solubility in water (approx. 0.1 %), but is highly soluble in organic solvents, such as alcohols, glycols and glycol ethers.

The solubility of sensiva™ sc 50 multifunctional in selected solvents is listed below. The solubility in the single solvents depends on the whole formulation and has to be tested in individual cases.

solvent	solubility [%]
Water	approx. 0.1
Ethanol	∞
Ethanol (10 % In water)	approx. 0.2
Propylene Glycol	approx. 30
Propylene Glycol (10 % in water)	approx. 0.2
Butylene Glycol	> 50
Butylene Glycol (10 % in water)	approx. 0.2
Paraffin Oil	∞
Silicone Oil	∞
Glycerin	approx. 1.0

general information

description of individual substances

ethylhexylglycerin	
	$C_{11}H_{24}O_3$ 204.31 g/mol
CAS no.	70445-33-9
CAS name	3-[(2-Ethylhexyl)oxy]-1,2-propandiol
CTFA name	Ethylhexylglycerin
ELINCS name	Sensiva SC 50
ELINCS no.	408-080-2
REACh registration no.	01-0000015745-65-0001

physical-chemical data	
appearance	clear, colorless – nearly colorless liquid
color index (Hazen)	max. 30
odor	slight characteristic
density (20 °C)	0.950 - 0.960 g/cm³
refractive index (20 °C)	1.449 - 1.453
pH-value (1.0 g/l, 20 °C)	6 – 8
vapour pressure (20 °C)	approx. 0.003 hPa
boiling point	145 °C (9 hPa)
solidification point	< -76 °C
flashpoint (ASTM D 93)	152 °C
water solubility (22 °C)	approx. 1.8 g/l
viscosity (Brookfield-RVT, 20 °C, spindle 1/20 Upm)	approx. 144 mPa s
distribution coefficient n-octanol/ water (20 °C, OECD 107)	log P_{ow} = 2.53

storage

Store at room temperature in the original container. Protect from heat and direct sunlight.

toxicology

Extensive toxicological tests confirm the good tolerability of sensiva™ sc 50 multifunctional. On oral administration and dermal application, sensiva™ sc 50 multifunctional proved to be non-toxic and non-irritant to the skin. The concentrate has an irritant effect on the ocular mucous membrane, but in a 5 % dilution sensiva™ sc 50 multifunctional does not irritate the eyes. In a test for a possible sensitising effect, no allergic reactions were observed. In in-vitro and in-vivo tests sensiva™ sc 50 multifunctional proved to be non-mutagenic. No phototoxic or photosensitising effects were found. The sensiva™ sc 50 multifunctional concentrate is labelled Xn-harmful (information in accordance with § 15a of the Chemicals Act). The results of all toxicological investigations are summarised in a toxicological statement which can be supplied upon request.

More details regarding handling, toxicity and labelling are available in the Safety Data Sheet (SDS).

[1] spreading coefficient has been determined by Cognis GmbH & Co. KG, Düsseldorf, Germany.
[2] sensory assessment test has been carried out at proDerm Institut für Angewandte Dermatologische Forschung, Hamburg, Germany.
[3] F. Heiss, Die Bestimmung des kosmetischen Gebrauchswertes von Riechstoffen mit Hilfe des Sniffing-Tests, Dragoco Report, 1976, 23. Jahrgang, 6, 131 – 147.

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